Assignment

Issue - 01 :
SQL operator Precedence

Location :
backend / src / main / java / com / internad / tasktracker / Task Repository . java
Also checked : db / queries / search - tasks . sql and Oracle reference package .

How I discovered it :
while testing search with status and archived records, I noticed that some results could bypass the status or archived condition.

Root cause :
SQL AND has higher precendence than OR .
The original condition was effectively evaluated as :
archived = FALSE AND title LIKE term OR description LIKE term AND status = status
Because of this, an archived task could match through its description, and the status condition was not ~~wotw~~ always applied to title matches.

Date

Fix :

I grouped the title and description conditions :

archived = FALSE AND (title LIKE term OR description LIKE term) AND status = status

I applied the same grouping to the oracle reference SQL.

Why this approach :

This was a small, low-risk fix that directly corrected the query logic without changing the ~~respo~~ repository design.

Issue 02 - Stale API responses
Location : use Tasks.js
Bug - multiple search requests could remain active at the same time.
fix : Added an isCancelled flag with use Effect clean up.
Result : Only the latest active request updates the UI.

Issue 03 - Loading state Bug.
Location - use Tasks.js
Bug : The UI remained stuck on "Loading..." when an API request failed.
Root cause : loading was not set to false inside the error handler.
fix : set loading = false on failure and clear old errors on new requests.
Result : API failures now display the proper error message.

Issue - 04 - Artificial API delay

location : Task Controller.java

Bug : The API had an unnecessary Thread.sleep() delay

Root cause : The controller artificially blocked requests.

Fix : Removed the Thread.sleep() logic

Result : API response time improved from around 1 second to about 10.67ms in testing.